Molecular-dynamics-based modal analysis of heat transport in quasi-one-dimensional systems from a symmetry-adapted perspective

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Modal decompositions of thermal conductivity from molecular dynamics (MD), such as Green–Kubo modal analysis (GKMA) and homogeneous nonequilibrium modal analysis (HNEMA), conventionally rely on the eigenvectors of the harmonic dynamical matrix. Within degenerate subspaces these eigenvectors are defined only up to an arbitrary unitary rotation, rendering per-mode conductivities implementation-dependent. We resolve this problem for quasi-one-dimensional (quasi-1D) systems by constructing the modal basis from line-group projection operators, so that every modal coordinate carries well-defined symmetry labels (k, m, Π_V) and the decomposition is unique at the level of irreducible representations (irreps). Applying the framework to a $(10, 0)$–$(20, 0)$ WS_2–MoS_2 double-walled nanotube (DWNT) described by a neuroevolution potential (NEP), we find at 300 K that the symmetry-resolved HNEMA and GKMA yield consistent conductivities, which are dominated by the E-type channels, with the largest contributions concentrated at low m and near $k = 0$. Eigenanalysis of the GKMA channel-pair matrix reveals that its leading eigenvalue carries 99.1% of the Frobenius spectral weight. The large pair terms therefore form one symmetry-resolved collective pattern rather than many independent contributions.

I. INTRODUCTION

Symmetry analysis has long been a cornerstone of how we understand the physical properties of crystalline solids, from the irrep labeling of phonon branches to the selection rules that govern phonon scattering and transmission [1–3]. In quasi-1D crystals such as nanotubes and nanowires, the relevant symmetry group is the line group, whose irreps are parameterized by quantum numbers such as the axial wave vector k, the angular-momentum index m and parity Π_V [4]. These quantum numbers label the vibrational modes and organize their inter-mode scattering and transmission channels.

In the interface regime, a symmetry-based organization of phonon transport has been put on a quantitative footing by atomistic Green's function methods [3]. That symmetry analysis was, however, restricted to the harmonic regime. At finite temperature, the instantaneous atomic configuration no longer respects the line group, although the equilibrium ensemble of the ideal structure still does. Extending symmetry analysis to finite-temperature MD therefore requires identifying how the observable transforms: selection rules for phonon amplitudes, scattering vertices, and modal heat-current correlations need not be the same.

MD-based thermal-transport methods are classified into equilibrium Green–Kubo MD, boundary-driven nonequilibrium MD, and homogeneous nonequilibrium MD frameworks [5, 6]. Modal and spectral decompositions exist in all of these settings: time-domain normal-mode analysis, spectral energy density, and GKMA are usually formulated in equilibrium MD, whereas direct-decomposition and spectral-heat-current approaches are often used in nonequilibrium MD. For the present symmetry question, however, the relevant subset is narrower: methods that decompose a homogeneous, periodic MD trajectory by projecting atomic velocities onto a complete modal basis. GKMA does this within the Green–Kubo framework, yielding a mode-pair conductivity matrix from modal heat-current correlations [7, 8]. HNEMA combines the same modal velocity projection with the efficiency of homogeneous nonequilibrium MD, giving diagonal per-mode conductivities that converge much faster, while sacrificing the mode–mode cross correlations accessible to GKMA [9, 10]. Spectral heat-current methods are highly useful for frequency-resolved transport [11, 12], but they do not by themselves provide the irrep-labelled modal pair correlations needed to resolve the symmetry structure of anharmonic heat transport.

Related roles for off-diagonal modal quantities arise in other formulations of thermal transport. The Allen–Feldman theory describes harmonic mode-coupling transport in disordered solids, QHGK expresses the conductivity as a mode-pair sum weighted by heat-flux matrix elements and linewidth overlap, and the Wigner formulation separates particle-like populations from wave-like interband coherences [13–15]. These quantities are not identical to the finite-temperature GKMA modal heat-current correlations considered here, but they establish a broader context in which off-diagonal contributions can become important for dense or quasi-degenerate vibrational spectra.

This modal projection viewpoint also exposes a gauge freedom that is usually hidden. The total heat current is recovered from the sum over a complete orthonormal

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basis, so the total thermal conductivity is invariant under unitary rotations of that basis. In a closely related frequency-domain normal-mode-analysis context, Feng et al. showed analytically and numerically that arbitrary orthonormal vectors can reproduce the same total spectral quantity, even though the individual branch-resolved contributions are redistributed [16]. The same issue appears in GKMA and HNEMA: per-mode conductivities are basis-dependent, whereas sums over invariant subspaces are physically meaningful. This becomes a fundamental problem in the presence of degeneracies, where numerical diagonalization returns an arbitrary unitary mixture within each degenerate subspace. Conventional eigenvectors therefore make the per-mode decomposition implementation-dependent and, more importantly for quasi-1D systems, leave the line-group quantum numbers of the transported heat current invisible.

In this work, we extend modal analysis to incorporate line-group symmetry from the outset. We replace the eigenvector basis used in conventional GKMA and HNEMA with a basis constructed from the line-group projectors, in which every modal eigenvector carries well-defined quantum numbers (k, m, Π_{V}). The resulting symmetry-projected HNEMA yields an irrep-resolved heat current and thermal conductivity. The line group of each structure and its full character table are obtained using the package Pulgon-tools [17]. We demonstrate the framework on the pristine $(10,0)$–$(20,0)$ WS_2–MoS_2 DWNT using MD trajectories driven by a NEP as implemented in GPUMD [18]. We first establish that the normal, symmetrized, and projector bases leave the total conductivity invariant, and then apply them to obtain the mode-decomposed thermal conductivity at 300 K. For the symmetrized and projector bases, we further resolve the contribution of each irrep.

The paper is organized as follows. Section II presents the methodology: we review the GKMA and HNEMA decompositions of the modal thermal conductivity (Secs. II A and II B), introduce the line-group projectors that generate a symmetry-adapted basis for quasi-1D systems (Sec. II C), formulate the resulting symmetry-projected modal analysis (Sec. II D), and give the computational details of the NEP training and MD protocol (Sec. II E). Section III reports the results: the accuracy of the trained potential (Sec. III A), the equivalence of the normal, symmetrized, and projector bases established on the WS_2–MoS_2 DWNT (Sec. III B), the irrep-resolved HNEMA transport at 300 K (Sec. III C), and the GKMA cross-term analysis (Sec. III D). Section IV summarizes the findings and outlines directions for future work.

II. METHODS

A. GKMA

The thermal conductivity tensor is written as the time integral of a modal heat-current autocorrelation function,

$$\kappa^{\alpha\beta}_{\lambda\lambda'} = \frac{1}{V_{\mathrm{eff}}\, k_B T^2} \int_0^\infty \left\langle Q^\alpha_\lambda(0)\, Q^\beta_{\lambda'}(t) \right\rangle \mathrm{d}t \tag{1}$$

where V_{eff} is the effective volume of the system, T is the temperature, k_B is the Boltzmann constant, α, β are Cartesian indices, and λ, λ' are mode indices.

The modal heat current is obtained by projecting the atomic velocities onto a modal basis $\{\mathbf{e}_\lambda\}$. The modal velocity coordinate is

$$v_\lambda(t) = \sum_i \sqrt{m_i}\, \mathbf{e}^*_{i,\lambda} \cdot \mathbf{v}_i(t), \tag{2}$$

where $\mathbf{e}_{i,\lambda}$ denotes the three-component block of $\mathbf{e}_\lambda$ on atom i, and the contribution of mode λ to the velocity of atom i is $\mathbf{v}_{i,\lambda}(t) = \mathbf{e}_{i,\lambda}\, v_\lambda(t)/\sqrt{m_i}$. The modal heat current then reads

$$Q^\alpha_\lambda(t) = \sum_i E_i\, v^\alpha_{i,\lambda}(t) + \sum_i \sum_{j\neq i} \left(-\frac{\partial \Phi_j}{\partial \mathbf{r}_i} \cdot \mathbf{v}_{i,\lambda}(t) \right) r^\alpha_{ij}, \tag{3}$$

where E_i is the total energy of atom i, Φ_j the potential energy assigned to atom j, and $\mathbf{r}_{ij}$ the position difference between atoms i and j. Summing Q^α_λ over the complete basis recovers the total heat current.

In conventional GKMA the modal basis $\{\boldsymbol{e}_\lambda\}$ is taken to be the set of eigenvectors of the harmonic dynamical matrix at the Γ point of the supercell. The construction of alternative symmetry-adapted bases is developed in Sec. II C and II D.

B. HNEMA

Introduced by Gabourie et al. [10], HNEMA decomposes the lattice thermal conductivity into per-mode contributions in a nonequilibrium MD framework. A small uniform driving force $\boldsymbol{F}_{\mathrm{e}}$ is applied to every atom along the transport direction, generating a steady-state heat current $\langle J^\alpha_\lambda(t)\rangle_{ne}$ in the linear-response regime. The mode-resolved thermal conductivity is obtained from:

$$\frac{\langle J^\alpha_\lambda(t)\rangle_{ne}}{T V_{\mathrm{eff}}} = \sum_\beta \kappa^{\alpha\beta}_\lambda F^\beta_{\mathrm{e}} \tag{4}$$

where $\kappa^{\alpha\beta}_\lambda$ is the thermal conductivity tensor of mode λ with Cartesian indices α, β, T is the system temperature

and V_{eff} is the effective volume. The modal heat current $J^{\alpha}_{\lambda}(t)$ is constructed exactly the same as Eq. 3 in GKMA.

In contrast to the per-mode-pair structure of Eq. 1, HNEMA yields a single-mode-index modal decomposition of thermal conductivity. Within the linear-response framework, the external driving force $\boldsymbol{F}_{\text{e}}$ induces a nonequilibrium heat current, and the resulting modal contribution is obtained from the projected modal heat current $\langle J^{\alpha}_{\lambda}(t)\rangle_{ne}$. Thus, HNEMA assigns a net conductivity contribution to each mode λ, but it does not explicitly resolve mode-mode cross correlations or a full $\kappa_{\lambda\lambda'}$ matrix. HNEMA is therefore complementary to GKMA, providing a statistically efficient route to single-mode conductivities that converges one to two orders of magnitude faster than the equilibrium Green–Kubo methods [10] while GKMA remains a natural tool for analyzing cross-mode contributions through Eq. 1. In the following sections we develop a symmetry-adapted construction of modal basis $\{\boldsymbol{e}_{\lambda}\}$ that applies to both methods.

C. Line-group symmetry and projectors for quasi-1D systems

A line group $L = Z \cdot P$ is the semidirect product of a generalized translational group Z, which may take the form of a pure translation, a screw axis $(C_Q \mid f)$ or a glide plane $(\sigma_V \mid a/2)$ and an axial point group P drawn from one of seven families (C_n, S_{2n}, C_{nh}, D_n, C_{nv}, D_{nd}, D_{nh}) [4]. Each irrep μ of the line group is parameterized by quantum numbers: the axial wave vector k, the angular-momentum index m and parity labels Π_{V}, Π_{H} or Π_{U}. Using the package Pulgon-tools [17], we identify the line group of input structures and obtain the character $\chi^{\mu}(g)$ for each line group element g in each irrep μ.

We construct the symmetry projector at wave vector k for irrep μ [19–21]:

$$\boldsymbol{P}^{\mu}(k) = \frac{d_{\mu}}{|G|}\sum_{g}\chi^{\mu}(g)^{*}\boldsymbol{S}(g) \tag{5}$$

where d_{μ} is the dimension of the irrep and $|G|$ is the order of the group. Because Eq. 5 is a character projector, it fixes each irrep subspace but leaves the partner labels within degenerate blocks up to unitary rotations; the irrep-block sums reported below are invariant under this remaining gauge freedom. The matrix $\boldsymbol{S}(g)$ acts on the $3N$-dimensional space of atomic displacements in the simulation box, and is defined by the displacement transformation matrix $\boldsymbol{M}(g)$ and two phase factors:

$$\boldsymbol{S}^{\alpha\beta}_{ij}(g) = \boldsymbol{M}^{\alpha\beta}_{ij}(g)\cdot e^{ikT_z}\cdot e^{ik(r^z_i - r^z_j)} \tag{6}$$

where $\boldsymbol{M}(g)$ is constructed by placing copies of the three-dimensional Cartesian rotation matrix associated with g at the rows and columns defined by the atom permutation induced by the symmetry operations. The first phase factor e^{ikT_z} is associated with the translational component of the group operation g, where T_z is the z component of the translational vector along the periodic axis. For integer translations this factor equals unity while for helical operations with fractional translations it becomes relevant. The second exponential factor $e^{ik(r^z_i - r^z_j)}$ carries the Bloch phase between the source and target atoms.

For each projector we extract an orthonormal basis for its image, $\boldsymbol{B}^{\mu}(k) = \left[\boldsymbol{b}^{\mu k}_{1},\ldots,\boldsymbol{b}^{\mu k}_{r_{\mu k}}\right]$, where $r_{\mu k} = \operatorname{rank}\boldsymbol{P}^{\mu}(k)$. In practice this basis is obtained from the column space of $\boldsymbol{P}^{\mu}(k)$, rather than by diagonalizing the projector as a dynamical matrix. Concatenating these bases across all irreps μ at a single wave vector k yields a symmetry-adapted basis that block-diagonalizes the dynamical matrix $\boldsymbol{D}(k)$ in the primitive cell.

GKMA and HNEMA are performed on a finite simulation supercell. When the simulation box contains K copies of the primitive cell along the periodic axis, the axial wave vectors are $k_n = 2\pi n/(Ka)$ with $n = 0, 1, ..., K-1$, where a is the primitive cell length. The supercell construction is performed separately for each allowed folded wave vector: for every pair (k_n, μ) we construct $\boldsymbol{P}^{\mu}(k_n)$, extract its column-space basis $\boldsymbol{B}^{\mu n}$, and only then concatenate the resulting bases. Thus the projectors for different k_n are not first combined into a single operator. The complete symmetry-adapted basis of the supercell is

$$|\text{SAB}\rangle = [\boldsymbol{B}^{\mu_1 0},\ldots,\boldsymbol{B}^{\mu_M 0},\boldsymbol{B}^{\mu_1 1},\ldots,\boldsymbol{B}^{\mu_M,K-1}], \tag{7}$$

where $\mu_1,\ldots,\mu_M$ denote the irreps present at a given k_n. Each column of $|\text{SAB}\rangle$ is a basis vector of the supercell displacement space carrying definite line-group quantum numbers (k_n, m, Π_{V}). We denote the λ-th column by $|\text{SAB}\rangle_{\lambda}$.

D. Symmetry-projector-based modal analysis

The modal analysis frameworks introduced in Sec. II A and II B are conventionally implemented using phonon eigenvectors obtained by diagonalizing the supercell dynamical matrix at Γ, for example with Phonopy [22, 23]. We now replace it with a basis adapted to the line-group symmetry of the supercell.

Starting from the symmetry-adapted basis $|\text{SAB}\rangle$ of Eq. 7, we consider two constructions of the modal basis $\boldsymbol{e}_{\lambda}$. The first is to use $|\text{SAB}\rangle$ directly. This projector basis is determined entirely by the line group and the supercell geometry, independent of the dynamical matrix or the interatomic potential. The second is to first transform the supercell Γ-point dynamical matrix into this basis, $\tilde{D} = \langle\text{SAB}|D|\text{SAB}\rangle$, whose blocks are labeled by the folded-wave-vector and irrep labels (k_n, μ), and then diagonalize each block separately. This yields symmetrized eigenvectors $\boldsymbol{e}^{\text{sym}}_{\lambda}$ that represent phonon modes carrying line-group quantum-number labels. The dynamical-matrix diagonalization is therefore performed after the

(k_n, μ) projector bases have been concatenated, and only within the symmetry blocks.

In both constructions, the modal basis is complete and orthonormal, so the modal heat currents Eq. 3 sum to the total heat current and the total thermal conductivity is preserved.

Replacing the eigenvector basis in Eq. 1 by either of the symmetry-adapted constructions yields an irrep-resolved thermal conductivity matrix in GKMA:

$$\kappa^{\alpha\beta}_{\mu,\nu} = \frac{1}{V_{\text{eff}}\, k_B T^2} \int_0^\infty \left\langle Q^\alpha_\mu(0) Q^\beta_\nu(t) \right\rangle \mathrm{d}t \tag{8}$$

where $Q^\alpha_\mu = \sum_{\lambda\in\mu} Q^\alpha_\lambda$ is the modal heat current summed over the modes of irrep μ. The diagonal terms $\kappa^{\alpha\beta}_{\mu\mu}$ quantify the conductivity carried by each symmetry channel, and the off-diagonal entries $\kappa^{\alpha\beta}_{\mu\nu}$ $(\mu \neq \nu)$ encode correlations between distinct channels.

There are two advantages of this construction compared with conventional GKMA. Firstly, each projector-basis vector belongs to a definite (k_n, μ) irrep subspace and therefore carries line-group quantum numbers regardless of dynamical-matrix degeneracy. Secondly, the projector basis itself requires no diagonalization of the harmonic dynamical matrix, since it is determined entirely by the symmetry of the structure; if phonon eigenvectors are desired, the subsequent diagonalization is confined to the symmetry blocks of $\tilde{D}$.

E. Computational details

All training labels were obtained from single-point density-functional-theory (DFT) calculations. We use the projector augmented-wave formalism [24] as implemented in the 6.5.1 release of VASP [25–27], together with the PBEsol approximation to the exchange and correlation energy [28] and the DFT-D3 method with a Becke–Johnson damping function [29] to approximately account for van der Waals interactions. We choose $5s^1 4d^5$, $6s^2 5d^4$ and $3s^2 3p^4$ as valence configurations for Mo, W and S respectively, a plane-wave cutoff energy of 600 eV, and $1 \times 1 \times 2$, $1 \times 1 \times 2$ and $2 \times 2 \times 1$ Γ-centred Monkhorst–Pack k-point grids for single-walled nanotubes, DWNT and 2D structures respectively. We moreover set an electronic convergence criterion of 10^{-6} eV and a Gaussian smearing of 0.01 eV for all single-point runs. For each configuration the total energy, atomic forces, and stress tensor were used as regression targets.

The interatomic potential is a locally descriptor-based NEP [18] natively implemented in GPUMD. NEP is well suited to the present work: it runs efficiently on GPUs, the heat-current decomposition required by GKMA and HNEMA is built in, and the modal basis used in that decomposition can be supplied externally, which is essential for the symmetry-adapted constructions of Sec. II D.

The reference configurations sample the WS_2 and MoS_2 monolayers, the pristine WS_2 $(10, 0)$ and MoS_2 $(20, 0)$ single-wall nanotubes, and the $(10, 0)$–$(20, 0)$ WS_2–MoS_2 DWNT. Finite-temperature snapshots at 300 K were drawn both from short *ab initio* MD trajectories and from foundation-model MD using the MACE-MATPES-PBE-0 foundation model [30, 31], and were augmented by elastically deformed structures to sample the strained regime. After single-point DFT labeling this yields a dataset of 920 configurations, partitioned into 820 training and 100 test structures. A single production potential was then trained on this dataset for 10^5 generations of the separable natural-evolution strategy, using cutoffs of 7 Å and 5 Å for the radial and angular descriptors respectively and including energy, force, and virial in the loss function. The resulting energy, force, and stress accuracies are reported in Sec. III A (Fig. 1) and are not repeated here.

All MD trajectories were propagated with this potential in GPUMD 4.2 [18], using a timestep of 1 fs. Unless otherwise stated, the production runs used the canonical (NVT) ensemble, with the temperature controlled by a Nosé–Hoover chain thermostat with a coupling time of 100 fs. Each trajectory was initialized with velocities drawn from a Maxwell–Boltzmann distribution at the target temperature and equilibrated in the NVT ensemble for 100 ps before the production run.

The application system is the pristine $(10, 0)$–$(20, 0)$ WS_2–MoS_2 DWNT, whose line group is $T'C_{10}$ (family 7) with a translational period of 180 atoms (40 Mo, 20 W, 120 S). All transport calculations use a $[1, 1, 10]$ simulation supercell, i.e. $K = 10$ primitive cells along the tube axis. The allowed axial wave vectors are therefore $k_n = 2\pi n/(10a)$ with $n = 0, \ldots, 9$, where a is the primitive-cell length. Following the family-7 convention of Table 4.7 in Ref. [4], the eight real-irrep channels are denoted $m0A = (0, -1)$, $m0B = (0, +1)$, $m1E = (1, 0)$, $m2E = (2, 0)$, $m3E = (3, 0)$, $m4E = (4, 0)$, $m5A = (5, -1)$, and $m5B = (5, +1)$, where each pair gives (m, Π_V). Thus A and B denote $\Pi_V = -1$ and $+1$, respectively, while E denotes the doubly degenerate channels with $m = 1$–4. Here V refers to the glide plane in the generalized translational group T', not to a vertical mirror in a C_{10v} point group; its one-dimensional characters contain the factor $\Pi_V e^{ika/2}$. Combined with the ten allowed wave vectors, these channels yield $8 \times 10 = 80$ irrep blocks that label the symmetry-adapted basis $|\mathrm{SAB}\rangle$ of the supercell.

Modal conductivities are reported per unit effective volume V_{eff} [Eq. 1]. We define $V_{\text{eff}} = \pi(R^2_{\text{out}} - R^2_{\text{in}})L$, where R_{in} and R_{out} are the minimum and maximum atomic radial distances from the tube axis; the annulus therefore extends from the inner surface of the inner wall to the outer surface of the outer wall, with no additional wall-thickness parameter. For the $[1, 1, 10]$ simulation supercell, $R_{\text{in}} = 3.925$ Å, $R_{\text{out}} = 12.235$ Å, and $L = 55.457$ Å, giving $V_{\text{eff}} = 2.3398 \times 10^4$ Å^3. This choice rescales all modal conductivities by a common factor and

therefore leaves the diagonal-to-cross-correlation ratios — the central quantities of Secs. III C and III D — unaffected.

For HNEMA, a small homogeneous driving force of magnitude 2×10^{-6} Å^{-1} is applied along the tube axis, weak enough to keep the system in the linear-response regime, and the steady-state modal heat currents are accumulated over the 50–100 ns window and averaged over 10 independent seeds (Sec. III C).

For GKMA, the irrep-resolved modal heat-current autocorrelation functions of Eq. 8 are evaluated from 100 ns equilibrium trajectories, and the resulting Green–Kubo running integral is accumulated as a function of correlation time up to 100 ns. For each seed, the running conductivity matrix is interpolated at 2 ns intervals over the 50–70 ns correlation-time window, where the running integral has plateaued (Fig. 10), and these values are averaged to obtain one conductivity matrix per seed. The reported matrix is the average of the six seed-resolved matrices. The reported total conductivity and its uncertainty are the mean and standard error, respectively, of the corresponding six window-averaged totals (Sec. III D).

III. RESULTS AND DISCUSSION

A. Accuracy of the NEP potential

The production NEP was trained on the 820-structure training set and validated on the 100-structure test set described in Sec. II E.

The final potential reaches root-mean-square errors of 2.61 and 2.68 meV atom^{-1} in energy and 53.1 and 54.7 meV Å^{-1} in force on the training and test sets, respectively, with the virial reproduced to 10.2 and 9.6 meV atom^{-1}. The parity plots of Fig. 1 show tight clustering along the diagonal for energies, forces, and stresses on both sets, confirming that the potential faithfully reproduces the DFT reference. This potential is used for all calculations reported below.

B. Comparison of the three modal bases

To examine the difference between these three bases (normal, symmetrized and projector), we compare the diagonal modal HNEMA thermal conductivity of the WS_2–MoS_2 DWNT under each of them. All three bases are evaluated on the same eight independent HNEMA runs, averaged over the common 10–40 ns diagnostic window, so that the three bases differ only in the choice of modal vectors and not in the underlying dynamics. The comparison shows how the modal contributions are redistributed among individual modes while the sum is preserved. This value refers to the short shared window and is not the converged conductivity reported in Sec. III C; the basis invariance holds independently of the averaging window.

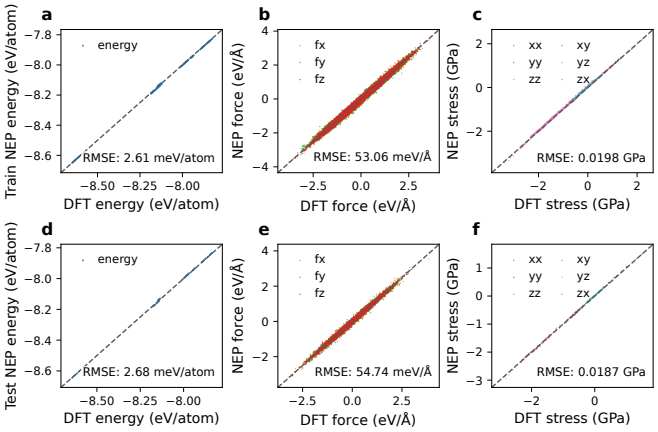


FIG. 1. The accuracy of the NEP results compared to the DFT ground truth. The comparison for the energies is given in (a) for the training set and in (d) for the test set. Analogous comparisons for the forces are shown in (b) and (e). Analogous comparisons for the stress are shown in (c) and (f).

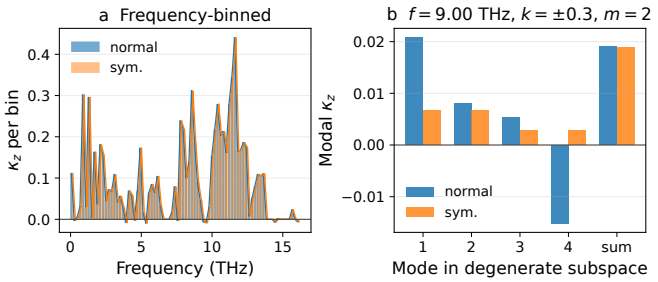


FIG. 2. Comparison of HNEMA diagonal modal axial thermal conductivity κ_z of the WS_2–MoS_2 DWNT between the normal and symmetrized bases. (a) Frequency-binned κ_z (80 bins spanning 0–16.2 THz), where each bar is the sum over all modes at the ten sampled axial wavevectors falling in that bin and the lines connect the bin values for the two bases. (b) Mode-resolved κ_z within a representative fourfold-degenerate subspace at $f = 8.99$ THz, angular-momentum quantum number $m = 2$ and $k = \pm 0.3\,(2\pi/a)$. The final bar is the sum over the four modes, which is invariant between the two bases despite the redistribution among the individual modes.

Figure 2 (a) shows the axial thermal conductivity in each frequency bin for the normal and symmetrized bases. The two distributions are in excellent agreement across the full frequency range, confirming that the frequency-resolved conductivity is unchanged by the symmetrization. The difference between the normal and symmetrized bases becomes apparent only at the level of individual degenerate modes. Figure 2 (b) shows a representative fourfold-degenerate subspace at 8.99 THz, formed by the doubly degenerate E branch with angular-momentum quantum number $m = 2$ evaluated at the symmetry-related wavevectors $k = \pm 0.3\,(2\pi/a)$. In the

normal basis, the modal conductivities are distributed unevenly among the four degenerate modes, and one of them even carries a spurious negative contribution. This behavior originates from the arbitrariness of the eigenvectors returned by the phonon diagonalization within a degenerate subspace, where any unitary rotation is equally valid, so that the normal basis is not uniquely defined and may vary with the numerical details of the diagonalization. In contrast, the symmetrized basis rotates the eigenvectors into symmetry-adapted combinations with well-defined irrep quantum-number labels, which distributes the conductivity evenly among the degenerate partners. As shown in Fig. 2 (b), the conductivity summed over the degenerate subspace is invariant between the two bases, while the individual modal contributions are redistributed into a unique symmetry-resolved form. The symmetrized basis therefore removes the arbitrary mixing between degenerate modes and provides a physically interpretable decomposition of the modal thermal conductivity.

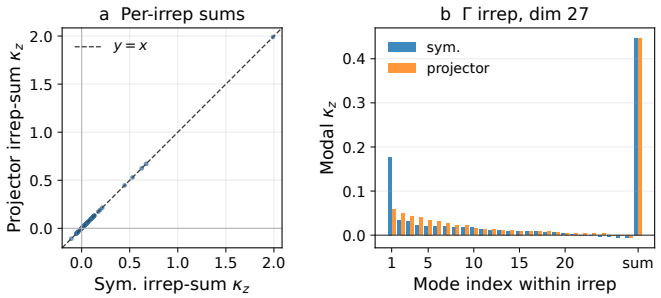


FIG. 3. Comparison of HNEMA diagonal modal axial thermal conductivity κ_z of the WS_2–MoS_2 DWNT obtained using the symmetrized eigenvectors and the projector-based basis. (a) Per-irrep summed κ_z; each of the 80 points corresponds to one line-group irrep and the dashed line indicates perfect agreement between the two bases. (b) Mode-resolved contributions within the Γ-point irrep with quantum numbers $(k, m, \Pi_V) = (0, 0, +1)$ and dimension 27. The final bar is the sum over the modes of that irrep.

We further compare the symmetrized basis with the projector-based basis in Fig. 3. Figure 3 (a) presents the thermal conductivity summed over all the modes belonging to the same irrep, for each of the 80 irreps that span the 5400 modes of the DWNT. The data points lie on the diagonal line, indicating that the irrep-resolved summed conductivity is identical for the two bases. This demonstrates that the physically meaningful conductivity associated with each irrep is invariant under different basis constructions inside the same symmetry subspace. The symmetrized and projector bases therefore span identical irrep subspaces and differ only by a unitary rotation within each block.

Figure 3(b) illustrates the resulting mode-level basis dependence within the Γ-point irrep $(k, m, \Pi_V) = (0, 0, +1)$. Although its summed conductivity within the same irrep is unchanged, their individual contribution are different. The porjector basis is constructed solely from the line-group symmetry and supercell geometry and therefore contains no material-specific dynamical information. It yields the same summed conductivity for each irrep as the symmetrized-eigenvector basis, but its individual vectors are not related with the real phonon modes.

C. HNEMA transport in the DWNT

We now apply the symmetry-projected HNEMA to the pristine $(10, 0)$–$(20, 0)$ WS_2–MoS_2 DWNT, whose line group is $T'C_{10}$ with eight (m, Π_V) symmetry channels. The calculations use the symmetrized-eigenvector basis, the $[1, 1, 10]$ supercell, and the driven steady-state protocol of Sec. II E. The seed-averaged total conductivity, evaluated over the 50–100 ns window, is $44.6\,\mathrm{W\,m^{-1}\,K^{-1}}$ at 300 K (10 seeds), with a seed-to-seed standard error of the mean of $15.5\,\mathrm{W\,m^{-1}\,K^{-1}}$; its seed convergence is shown in Fig. 9. The irrep-resolved HNEMA conductivity matrix is strictly diagonal: HNEMA assigns one conductivity contribution to each projected mode and does not construct a mode-pair matrix.

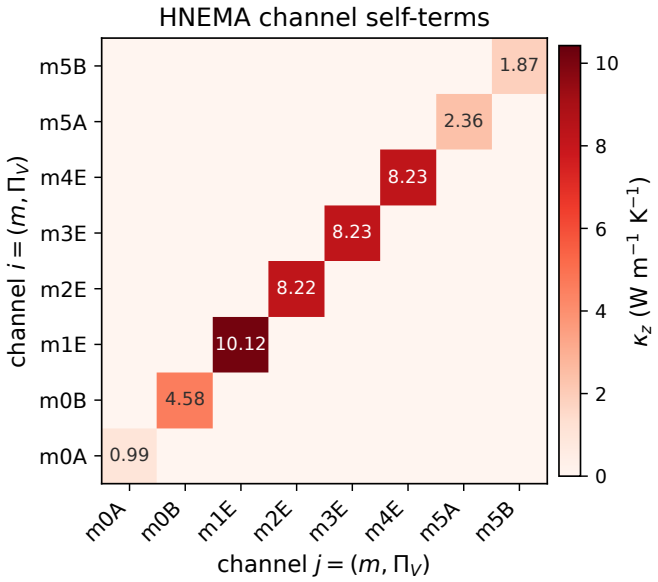


FIG. 4. Per-channel HNEMA thermal conductivity of the DWNT at 300 K, shown as the 8×8 (m, Π_V) matrix summed over the axial wave vector k. Because HNEMA is diagonal by construction, the off-diagonal blocks are empty.

Summing the diagonal contributions over the axial wave vector within each symmetry channel shows that transport is dominated by the low angular-momentum channels. The doubly degenerate E-type channels $m = 1$–4 carry the bulk of the conductivity, with $m = 1$ the single largest contributor, while the one-dimensional $m = 0$ and $m = 5$ channels contribute least as shown in Fig. 4. Resolving all eight (m, Π_V) channels in k fur-

ther shows that the conductivity is concentrated near $k \approx 0$, where the long-wavelength, high-group-velocity modes reside, and this concentration is most pronounced for the low-m channels as shown in Fig. 5.

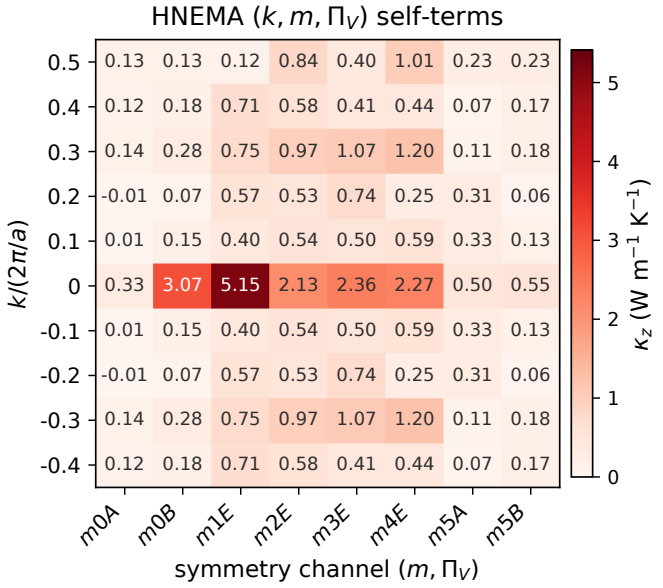


FIG. 5. (k, m, Π_V)-resolved HNEMA thermal conductivity of the DWNT at 300 K, with the axial wave vector k (in units of $2\pi/a$) on the vertical axis and the eight symmetry channels on the horizontal axis. For the one-dimensional channels at $m = 0$ and 5, A and B denote $\Pi_V = -1$ and $+1$, respectively, while $m = 1$–4 are doubly degenerate E-type channels. Each cell gives the corresponding contribution in W m^{-1} K^{-1}.

A small number of weakly contributing modes yield slightly negative modal conductivities. These values lie well within the seed-to-seed SEM and represent statistical noise rather than physical heat flow against the gradient; they have a negligible effect on the channel sums and on the total.

D. GKMA analysis in the DWNT

We now turn to the equilibrium GKMA decomposition of the same DWNT, which, unlike HNEMA, retains the full mode-pair structure of the conductivity. Averaging the GKMA running conductivity over the 50–70 ns correlation-time window for each trajectory and then over six independent seeds gives a total axial (z) conductivity of 45.3 W m^{-1} K^{-1}, with a seed-to-seed standard error of 11.7 W m^{-1} K^{-1} for the per-seed window averages. The GKMA total agrees with the HNEMA value at 300 K (44.6 W m^{-1} K^{-1}, with a standard error of 15.5 W m^{-1} K^{-1}; Sec. III C), well inside the statistical uncertainty of either estimate. Since GKMA carries the full off-diagonal cross-term structure while HNEMA folds every contribution onto the diagonal, their agreement on the total is a clean cross-check that the two independent decompositions recover the same conductivity. The convergence of the GKMA total with correlation time is shown in Appendix A Fig. 10.

Unlike the strictly diagonal HNEMA decomposition of Sec. III C, the full GKMA irrep-pair conductivity matrix is dominated by its off-diagonal entries. The $[1, 1, 10]$ simulation supercell used in the MD calculations contains ten primitive cells along the tube axis and therefore yields ten allowed axial wave vectors. Combined with the eight (m, Π_V) symmetry channels, these wave vectors give the full 80×80 irrep-pair conductivity matrix. After summing over both axial-wave-vector indices, this matrix reduces to the 8×8 symmetry-channel-resolved GKMA thermal-conductivity matrix shown in Fig. 6. Its off-diagonal entries quantify the contributions to the total conductivity arising from heat-current cross-correlations between distinct (m, Π_V) channels. The underlying 80×80 matrix is shown in Appendix A (Fig. 11).

Quantitatively, partitioning the total into (i) within-block contributions (same k, m, Π_V), (ii) same-channel (cross-k) correlations (same (m, Π_V) but different k), and (iii) cross-channel correlations (different (m, Π_V)) gives contributions of 6.3%, 14.2%, and 79.4%, respectively, with jackknife standard errors of 1.9, 2.0, and 3.2 percentage points.

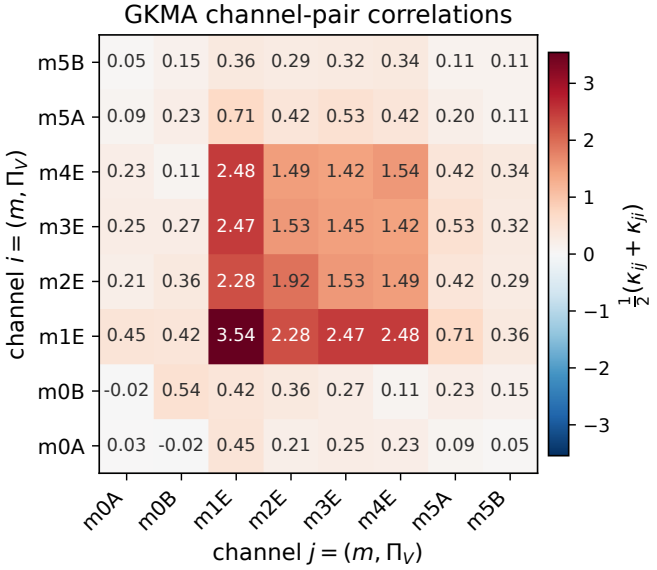


FIG. 6. Symmetry-channel-resolved GKMA 8×8 (m, Π_V) thermal-conductivity matrix of the DWNT at 300 K, obtained by summing the full matrix over both axial-wave-vector indices and averaging over 6 seeds. Entries are the symmetrized values $\frac{1}{2}(\kappa_{IJ} + \kappa_{JI})$ in W m^{-1} K^{-1}. For the one-dimensional channels at $m = 0$ and 5, A and B denote $\Pi_V = -1$ and $+1$, respectively; $m = 1$–4 are doubly degenerate E-type channels.

Before turning to the finite-temperature data, we clarify the symmetry of the pair matrix itself, which motivates the symmetrized display of Fig. 6. The modal conductivity of Eq. 8 is a one-sided time integral, $\kappa_{IJ} \propto$

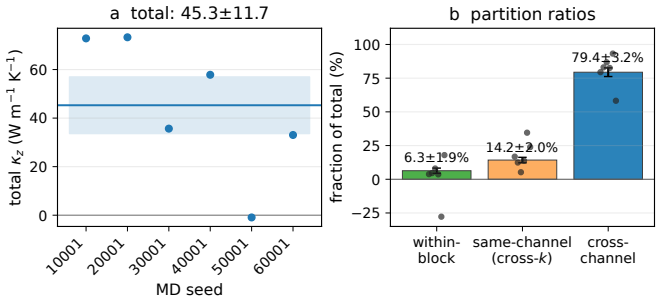


FIG. 7. Per-seed stability of the GKMA partition at 300 K. (a) Total axial conductivity for each of the 6 seeds (points), with the seed mean (45.3 W m^{-1} K^{-1}) and its standard error (11.7 W m^{-1} K^{-1}) shown as the band. (b) Three-way partition into within-block, same-channel (cross-k), and cross-channel contributions. The error bars are the ratio-of-sums estimate with jackknife standard errors, and the overlaid points are the per-seed fractions.

$\int_0^\infty C_{IJ}(t)\,dt$ with $C_{IJ}(t) = \langle Q_I(0)\,Q_J(t)\rangle$. The naive expectation that this matrix be symmetric assumes $C_{IJ}(t) = C_{JI}(t)$, but stationarity of the equilibrium ensemble gives instead a time-reversal of the lag,

$$C_{IJ}(t) = \langle Q_I(0)Q_J(t)\rangle = \langle Q_I(-t)Q_J(0)\rangle = C_{JI}(-t). \tag{9}$$

Consequently $\kappa_{JI} \propto \int_0^\infty C_{JI}(t)\,dt = \int_0^\infty C_{IJ}(-t)\,dt = \int_{-\infty}^0 C_{IJ}(t)\,dt$, the two one-sided integrals κ_{IJ} and κ_{JI} sample the positive- and negative-lag halves of the *same* cross-correlation function. The one-sided GKMA conductivity matrix is therefore generally asymmetric, with $\kappa_{IJ} \neq \kappa_{JI}$. To represent each channel pair by a single value, we use the symmetrized matrix $\kappa_{IJ}^{\mathrm{sym}} = \frac{1}{2}(\kappa_{IJ} + \kappa_{JI})$ for presentation and the subsequent channel-pair analysis. This symmetrization preserves both the total conductivity and the partition sums because the two mirrored cells together retain the full pair contribution $\kappa_{IJ} + \kappa_{JI}$.

At 300 K, the cross-channel blocks carry the majority of the measured conductivity (Fig. 6 and Fig. 7). The weight is concentrated in the doubly degenerate E-type channels ($1 \le |m| \le 4$, without Π_V), whose mutual couplings account for 23.4 W m^{-1} K^{-1} (52% of the total). The largest individual couplings all involve $m1E$, which behaves as a coupling hub. The k-resolved matrix (Appendix A, Fig. 11) further shows that correlations touching k_0 (Γ) make up half of all off-diagonal weight.

This structure becomes clearer by diagonalizing the seed-averaged symmetrized channel-pair matrix K. Writing

$$K = U\Lambda U^{\mathrm{T}}, \qquad \kappa_z = \mathbf{1}^{\mathrm{T}}K\mathbf{1} = \sum_\alpha \lambda_\alpha \left(\boldsymbol{u}_\alpha^{\mathrm{T}}\mathbf{1}\right)^2, \tag{10}$$

Here, $\boldsymbol{u}_\alpha$ is an eigenvector of K in the eight-dimensional symmetry-channel space and should not be confused with the mode-resolved basis vector $\boldsymbol{e}_\lambda$ in the $3N$-dimensional displacement space. The seed-averaged matrix is nearly rank one: its leading eigenvalue accounts for 99.1% of the Frobenius spectral weight, with a jackknife standard error of 0.7 percentage points. The near-rank-one form, $K \approx \lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^{\mathrm{T}}$, shows that the large pair terms form a single collective transport pattern: the participating symmetry-resolved irrep channels are correlated components of a common channel-space structure rather than independent pairwise contributions. The leading direction is distributed across several E-type channels, with $m1E$ carrying the largest component (Fig. 8).

To test whether this collective channel-space structure is an artifact of the production thermostat, we additionally performed three 20 ns NVE trajectories after the same 100 ps NVT equilibration. Modal heat currents were sampled every 10 fs, and the conductivity matrices were averaged over the 5–10 ns correlation-time window. The three NVE matrices were then averaged and subjected to the same symmetrization and eigendecomposition as the NVT data. The NVE-averaged matrix is likewise nearly rank one, with its leading eigenvalue accounting for 99.5% of the Frobenius spectral weight (three-seed jackknife standard error: 0.7 percentage points), and its leading eigenvector reproduces the same E-type-dominated profile, with $m1E$ carrying the largest component [Fig. 8(b)].

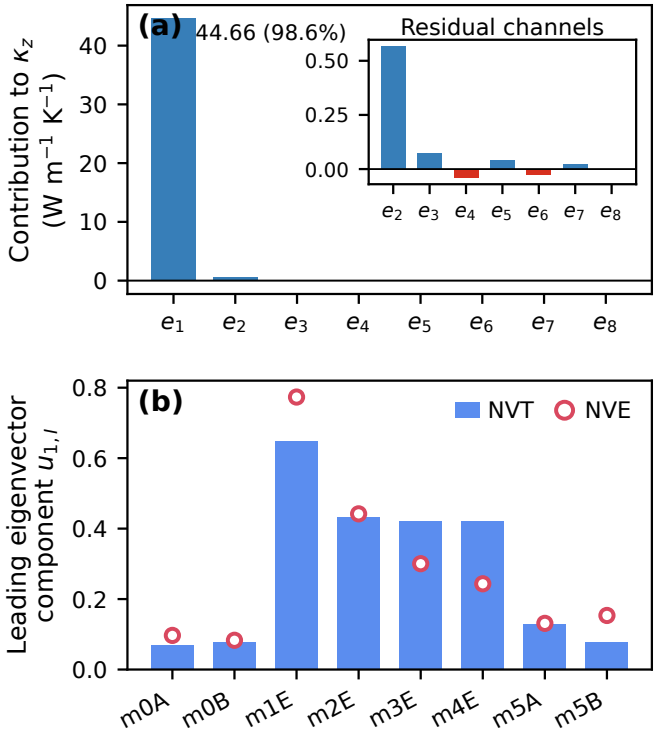


FIG. 8. Collective eigenchannels of the seed-averaged symmetrized 8 × 8 GKMA matrix at 300 K. (a) Signed conductivity contributions $\lambda_\alpha(\boldsymbol{u}_\alpha^{\mathrm{T}}\mathbf{1})^2$, ordered by decreasing absolute magnitude. The inset expands the seven residual channels. (b) Components of the leading eigenvector for the 6-seed NVT average (bars) and the independent 3-seed NVE average (open circles).

IV. CONCLUSION

We have extended MD-based modal analysis to incorporate line-group symmetry from the outset, replacing the conventional eigenvector basis with a symmetry-adapted basis in which every modal coordinate carries definite line-group quantum numbers (k, m, Π_V). Within this basis both GKMA and HNEMA yield a unique, irrep-resolved thermal conductivity, free of the arbitrary unitary mixing that afflicts degenerate eigenvectors in conventional implementations. The two methods are complementary: HNEMA provides a statistically efficient, diagonal-by-construction decomposition of the per-channel conductivity, while GKMA recovers the full irrep-pair matrix, including the off-diagonal cross-channel correlations that HNEMA cannot access. On the WS_2–MoS_2 DWNT we verified that the symmetrized and projector constructions leave the total conductivity invariant, and that GKMA and HNEMA agree on the total to within the seed-to-seed uncertainty, confirming that the symmetry-adapted decomposition preserves the sum rule.

Applying the framework to the pristine $(10, 0)$–$(20, 0)$ WS_2–MoS_2 DWNT, we find substantial off-diagonal heat-current correlations between distinct family-7 (m, Π_V) channels at 300 K, concentrated in the low-k, low-m irreps. The leading eigenvalue of the seed-averaged symmetrized matrix carries 99.1% of its Frobenius spectral weight, showing that the pair terms form a collective transport pattern rather than independent contributions. Its eigenvector is concentrated in the E-type channels, and its dominant $m1E$ component accounts for the hub pattern in the channel-pair matrix. An independent NVE control reproduces this collective direction, showing that the organized near-rank-one structure is robust to removal of the production thermostat.

These results establish a symmetry-adapted framework for resolving thermal transport by line-group irreps while retaining the cross-channel correlation structure of the modal heat current.

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Appendix A: Supplemental figures

This appendix collects supplemental figures: the seed convergence of the HNEMA total at 300 K (Fig. 9), the correlation-time convergence of the GKMA total (Fig. 10), and the full 80×80 irrep-pair conductivity matrices resolved in the axial wave vector (Fig. 11).

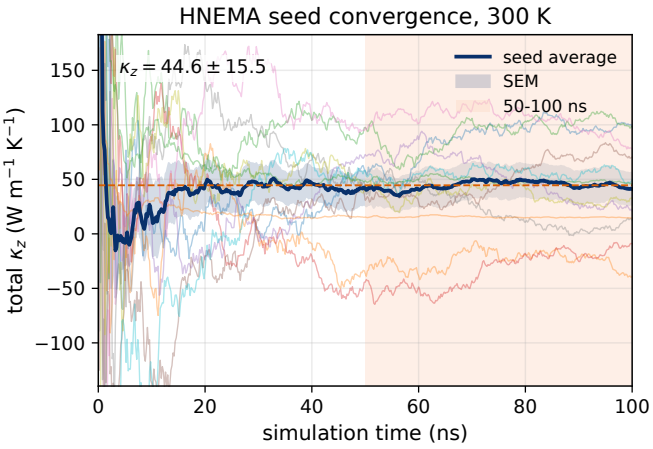


FIG. 9. Seed convergence of the total HNEMA conductivity of the DWNT at 300 K, averaged over the 50–100 ns window.

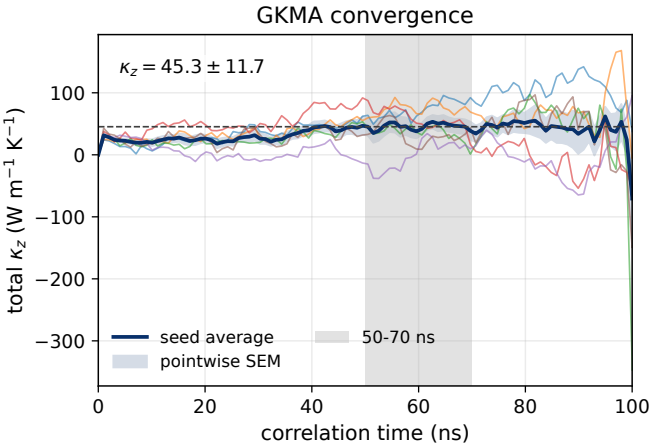


FIG. 10. Convergence of the total GKMA axial conductivity of the DWNT at 300 K with correlation time (6 seeds). The reported mean is 45.3 W m^{-1} K^{-1}, with a seed-to-seed standard error of 11.7 W m^{-1} K^{-1}. Each seed value is obtained by interpolation at 2 ns intervals over the 50–70 ns window to match the pair-matrix averaging; the shaded band shows the pointwise standard error.

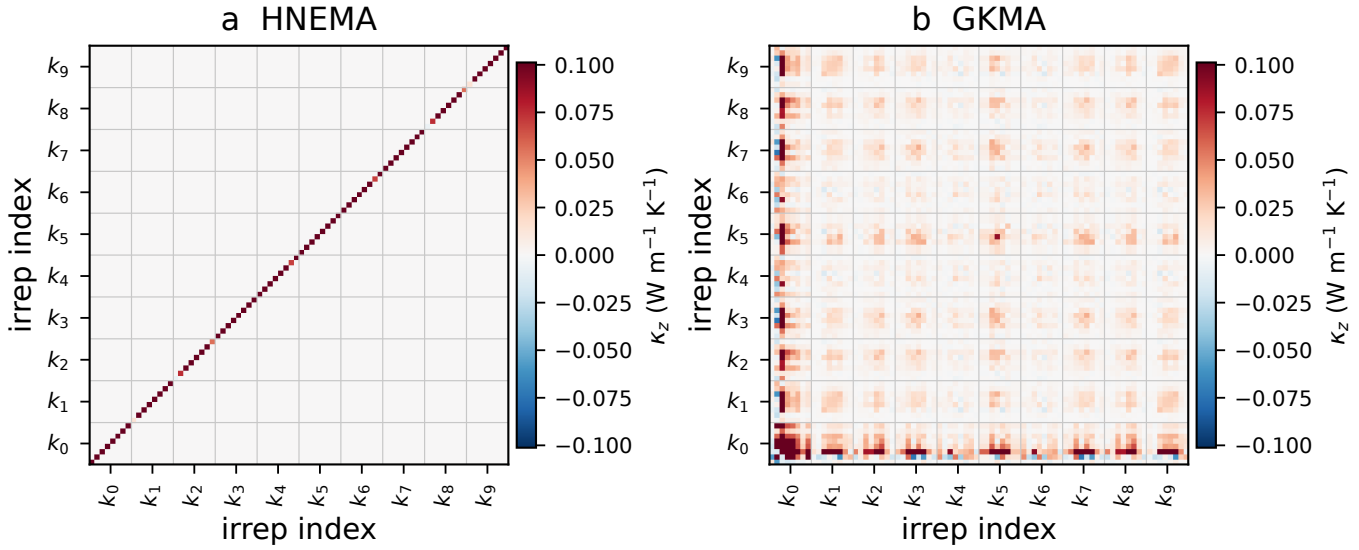


FIG. 11. Axial-wave-vector-resolved full 80×80 irrep-pair thermal-conductivity matrices of the DWNT at 300 K (eight (m, Π_V) channels $\times$ ten axial wave vectors $k_0 \ldots k_9$; grey lines separate the k blocks). (a) HNEMA, which is diagonal by construction. (b) GKMA, which exposes the off-diagonal correlations between modal heat-current components that are inaccessible to HNEMA. The GKMA matrix is shown symmetrized for display, $\frac{1}{2}(\kappa_{IJ}+\kappa_{JI})$; the physical pair contribution is $\kappa_{IJ}+\kappa_{JI}$. The off-diagonal texture is overwhelmingly positive (red) and organized around the k_0 (Γ) row and column, which acts as a coupling hub. Both matrices are averaged over 6 seeds in the 50–70 ns window.

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